PREDICTION OF FETAL HEART RATE FROM CARDIOTOCOGRAPHY SIGNALS: THE USE OF MACHINE LEARNING ON UCI CTG DATASET

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1. SUMMARY

Cardiotocography has been established as the cornerstone of intrapartum fetal and antenatal surveillance, however, interpretation has suffered from substantial inter and intra-observer variability. In this study, machine learning techniques are applied to the UCI cardiotocography dataset (2,126 samples and 21 features), with a 3-class label: **Normal/Suspect/Pathologic**. Exploratory analysis was explored in this study, after which the data was split into stratified training and test partitions. Z-score was used to standardise the features using parameters fitted on training alone and class imbalance was addressed using SMOTE. Feature selection methods including **Mutual Information, RFE and LASSO** were used to derive an **11-feature** consensus subset. Four classifiers were compared in this study: Logistic Regression, SVM, Random Forest and XGBoost and was tuned with a stratified 5-fold cross-validation technique and assessed under the three-problem framing; 3-class, binary and reduced feature 3-class. **XGBoos**t achieved **94.6% accuracy** and **AUC = 0.982** in the **3-class** setting whilst in the binary model **XGBoost** reached **accuracy 95.5%** and **AUC = 0.988**, both being the best model in each category. **ASTV, ALTV, MSTV and FHR central tendency (Mean**) emerged as the most predictive features, in close alignment with FIGO 2015 clinical criteria.

2. INTRODUCTION

The use of cardiotocography in continuous fetal heart rate and uterine contraction monitoring is the most widely assessable method of monitoring fetal wellbeing during the late phase of pregnancy and labour. Irrespective of this, CTG clinical drawback is a comparatively poor positive predictive value together with significant intra- and inter observer disagreement [1]. The risk of fetal hypoxia, neurological injury, under-treatment and over treatment is one of the potential risks the misinterpretation brings to this field. The use of computer assisted decision support has been ongoing research for over three decades, starting with rule-based systems like SisPorto that was encoded with FIGO guidelines into predictable algorithms [2].

Machine learning (ML) offers an alternative data-driven route. ML algorithms learn the mapping of the CTG features to fetal state directly from labelled data instead of hand coding the clinical rules. In a recent review by Francis *et al*. [3], it was found that a wide variety of ML approaches have been applied to CTG dataset, however, comparing across different studies is disrupted by inconsistent feature sets, target definitions and evaluation metrics. Francis et al. Specifically state that pipelines are not reproduceable and detailed reporting of the performance under class imbalance is the hallmark of CTG dataset where the pathological cases are rare.

This report addresses the concrete questions of the brief: (i) can the 21 quantitative CTG features predict fetal state with performance materially above chance? (ii) which features are the most indicative of abnormal states; and (iii) is there a reduced feature subset that retains strong predictive performance? In addition to these core questions, the robustness by training a binary model on the extreme classes (Normal vs Pathologic) was explored and applying these suspect cases to it, this is a strategy designed to specifically expose if the Suspect label is an accurate intermediate state or a heterogeneous mixture.

3. LITERATURE REVIEW

3.1 FIGO Framework and Clinical Context

FIGO 2015 consensus guidelines govern the interpretation of modern CTG, which classifies a tracing on five features; baseline FHR, baseline variability, presence of accelerations, presence and type of decelerations, and uterine activity [1]. A normal baseline is between 110 and 160 bpm; prolonged and decreased baseline variability (usually below 5bpm bandwidth amplitude over prolonged durations) which are the main indicators of potential fetal impairment. The 21 features in the UCI CTG dataset are based on the same physiological construct, for instance, FIGO variability criteria is directly encoded by ASTV (abnormal short-term variability) and ALTV (abnormal long-term variability), while DL, DS and DP represent light, severe and prolonged decelerations, respectively.

SisPorto generated the dataset itself, which is the longest established computerised CTG analyser, that produces FIGO-aligned features automatically [2], [4]. Three expert obstetricians assigned the three-class fetal-state label (Normal, Suspect, Pathologic) from their observations, which provided a high-quality but non-trivial gold standard label.

3.2. Machine Learning Approaches.

In a recent review by Francis et al. [3], ML CTG literature was synthesized into a single scoping review, the key findings of the study were that classical tabular learners including; random forest, gradient boosting and support vector machines dominate published works, although deep models trained on raw FHR signals perform competitively, they require a larger dataset and that prevents comparability across papers by inconsistent reporting standards and benchmarks. Three persistent methodological gaps were identified in the reviews: (i)feature selection procedure that were either absent or not sufficiently justified, (ii) inadequate handling of the severe class imbalance observed, with other studies demonstrating accuracy along on majority skewed test sets and (iii) limitation of robustness analysis of the ambiguous suspect class [3]. Other studies have shown large-scale efforts and nationwide multicentre CTG model to demonstrate that deep neural networks trained on hundreds of thousands of person-minutes can match or even exceed expert performance [5], [6], however not at a scale available with the UCI dataset.

Tabular CTG data specifically have several published studies using the same UCI corpus [7], [8], [9], these studies have all reported binary or three class accuracies in 90 –96% range with the use of ensemble methods that is broadly in line with the figures reported in this study. They acknowledged limitations such as; failure to apply resampling on the training set only, which led to optimistic evaluation on the test-set also the lack of feature importance interpretation in grounded clinical physiology. This study addresses both concerns stated as SMOTE [10] is applied strictly in the training partition and feature importance is presented with the FIGO physiological interpretation.

3.3. Gaps Addressed In this Analysis

The triangulation of Mutual Information, Recursive Feature Elimination [11] and L1-LASSO [12] methodologically provides a robust feature ranking that does not rely on a single feature algorithm which helps remove bias. Secondly, comparing with four diverse classifiers under the similar preprocessing and identical metrics such as; accuracy, macro F1, sensitivity, specificity and ROC-AUC, we were able to provide a fair study specific benchmark. Lastly, training Normal vs Pathologic model and its application to Suspect cases, we were able to directly assess if the Suspect label is a significant intermediate or simply heterogenous, this is an analysis that is rarely seen in published literature on this specific dataset.

4. METHODOLOGY

4.1. Dataset and Preprocessing

The UCI cardiotocography dataset records 2,126 samples, each described by 21 numerical features gotten from FHR and Uterine Contraction tracings, with a 3-class fetal-state label [13]. The dataset was loaded focusing on the specified 21 features and NSP target was retained. A strong class imbalance was observed (Normal 77.8%, Suspect 13.9%, Pathologic 8.3%), the binary version of the target was constructed by the combination of the Suspect and Pathologic into a single Abnormal class.

To preserve the class proportions, the data was split into stratified training and test partitions of 80/20. Continuous features were standardized by the removal of the mean and scaling to unit variance with the aid of parameters fitted on the training set only, this was in line with the best practice for SVM and logistic regression.

4.2. Class Imbalance

The use of Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training partition only, with the k = 3 nearest neighbours being given the small absolute size of the Pathologic class. This method prevents duplication of samples by generating synthetic minority samples by linear combination between existing minority points and the k nearest neighbours, further increasing the decision region of the minority class [10]. Each of the three classes contained 1,323 training samples after resampling. Importantly, the test set was left untouched to allow all reported test metrics reflect its performance on the natural class distribution.

4.3. Feature Selection

Mutual Information; a model-free measure of statistical dependence between each feature and target was implemented in this study via ***mutual_info_classif*** in ***scikit-learn*** [14]. Recursive Feature Elimination (RFE) that wraps a Random Forest base estimator and removes the least important feature until a subset of 10 is attained was another feature selection method explored in this study [11]. Lastly, LASSO removed weakly informative coefficients to zero and then performs embedded selection [12]. These were the three feature selection method used in this study and a consensus yielded 11 features.

4.4. Classifiers and Hyperparameter Tuning

To avoid relying on a single algorithm's assumptions, four classifiers each with a different inductive bias was trained on the SMOTE balanced training set: the multinomial Logistic Regression with L2 Regularisation, RBF-kernel Support Vector Machine, Random Forest [15], XGBoost gradient-boosted trees [16]. Each of these models were tuned by a stratified 5-fold cross validation on the training set. The grid included C ∈ {0.01, 0.1, 1, 10} for Logistic Regression and SVM, *max_depth* ∈ {None, 10, 20} for Random Forest, and *n_estimators* $\in$ *{100, 200}* with *learning_rate* $\in$ *{0.05, 0.1}* for XGBoost. Macro-F1 was used as the cross-validation criterion to give equal weight to all three classes.

4.5. Evaluation

Accuracy, macro-averaged F1, macro-averaged sensitivity (recall), specificity and ROC-AUC were measured on the test set and computed in a comparison manner for the multiclass case. The use of confusion matrices was used to locate clinically significant errors. For instance, in a case of false-negative Pathologic predictions.

The three-problem framing was compared which included; a 3-class (Normal, Suspect, Pathologic), a binary (Normal vs Abnormal) both with all 21 features and 3-class with the 11-feature consensus subset.

Finally, to check for robustness, XGBoost model was trained on the extreme classes only (Normal and Pathologic) and further applied, which resulted to a pathology probability to **295** Suspect cases.

5. RESULTS AND DISCUSSION

5.1. Exploratory Data Analysis

As seen in Figure 1 below, strong class imbalance is observed in both the three-class and binary framings. The pathologic class is about 8.3% of the records, motivating the use of SMOTE on the training partition.

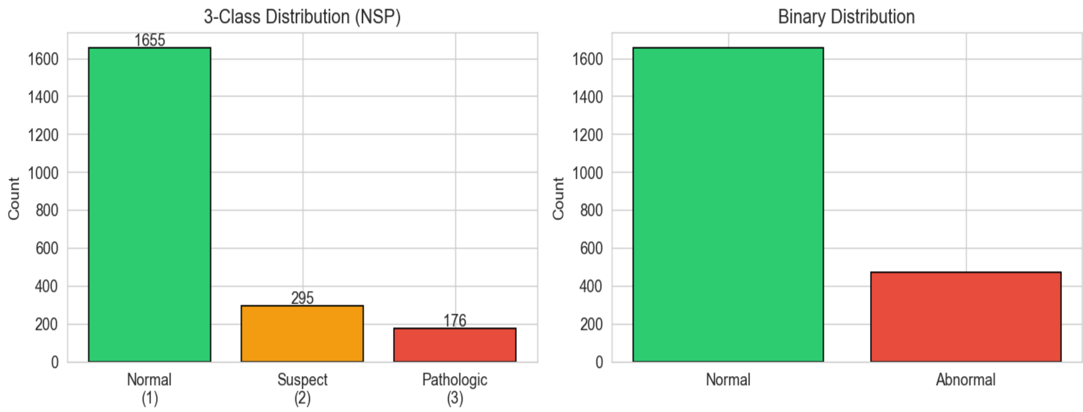

Figure 1: Class distribution for the three-class (NSP) and binary (Normal vs Abnormal) target.

Figure 2 shows the inter-feature correlation structure. There are two distinct blocks that appear: the histogram-shape block (Width, Min, Max, Variance) and the strongly inter-correlated central tendency blocks (Mode, Mean, Median, LB) with $\boldsymbol{p} > \mathbf{0.9}$ in some pairs. Redundancy is the clearest reason for feature selection; any model trained on all 21 features must explicitly learn to disregard the duplicate signals.

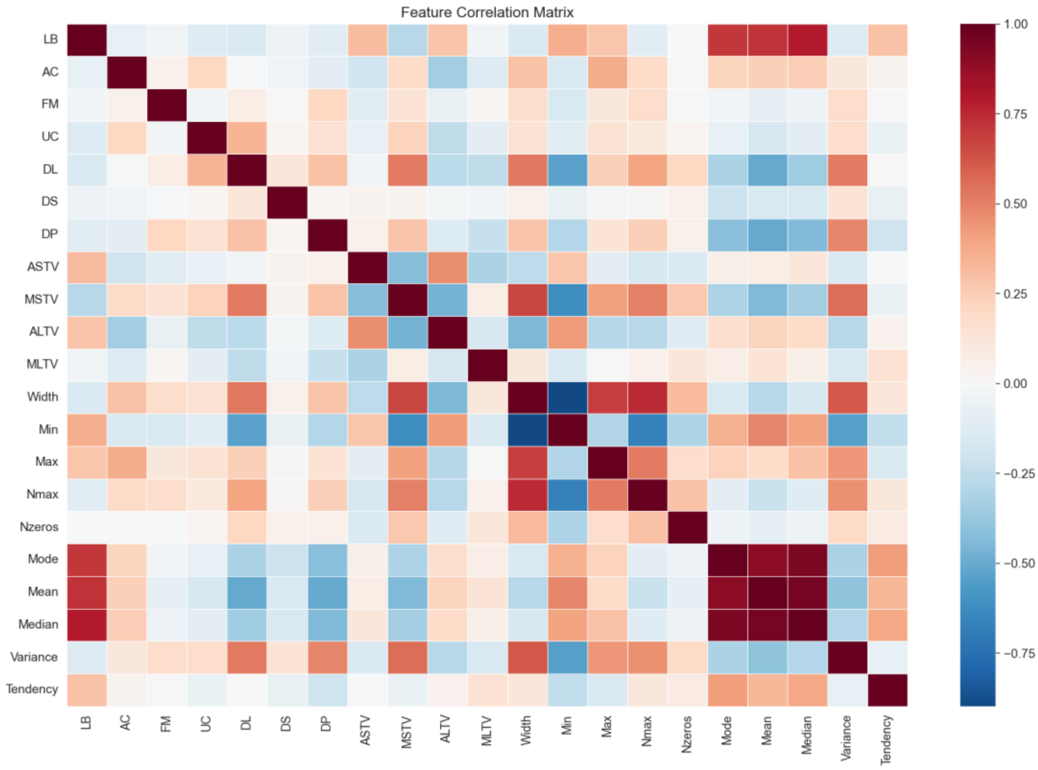

Figure 2: Pearson correlation matrix of the 21 quantitative CTG feature, showing visible intercorrelated blocks.

The class conditional histograms of representative features in Figure 3 below reveals that single features e.g.: LHR (baseline FHR) and AC (number of accelerations) do not separate class cleanly. Pathologic cases are seen at the tails of the FHR distribution, and they have very few accelerations, however the overlap in Normal case is substantial. Since no single feature separates the classes cleanly, the multivariate, ensemble-bases modelling approach taken in this study is supported.

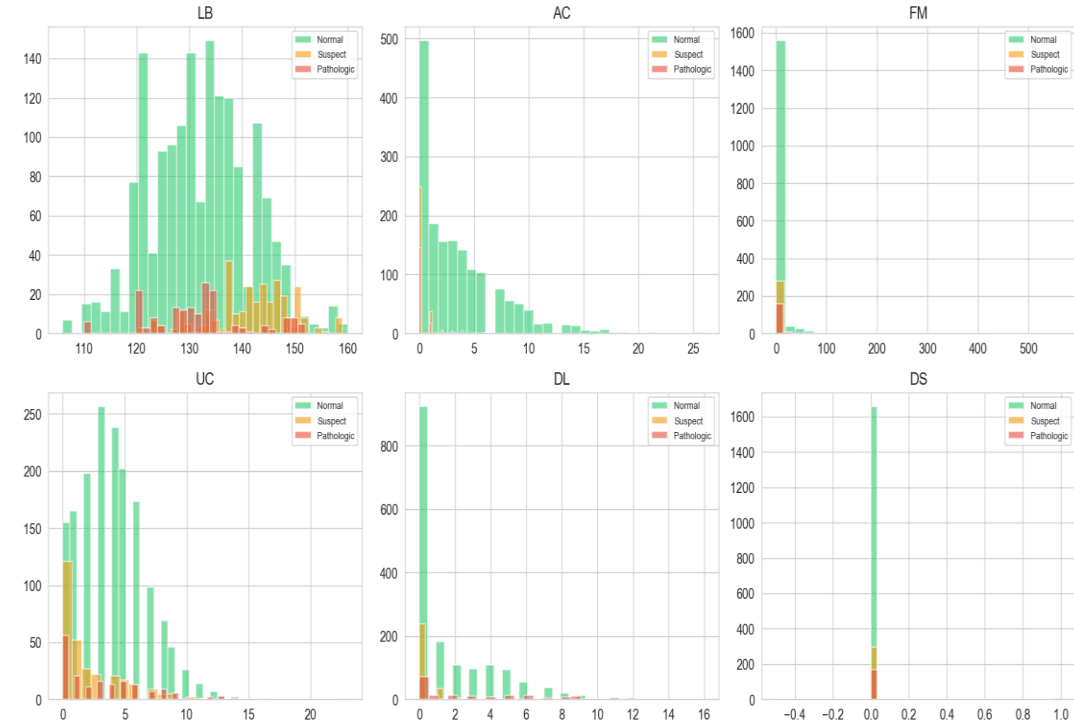

Figure 3: class conditional histograms of six representative features, showing that no single feature is diagnostic.

5.2. Feature Selection

Mutual Information, RFE, LASSO L1 were the feature selection methods explored in this study. The striking convergence observed in Mean, Median, MSTV, ASTV, MLTV, AC and LB as they appeared in top 10 of all three methods. Figure 4 shows the top features of all three methods, 11 features were selected by at least two methods and then form the consensus subset: AC, ALTV, ASTV, DP, LB, MLTV, MSTV, Mean, Median, Mode, Variance.

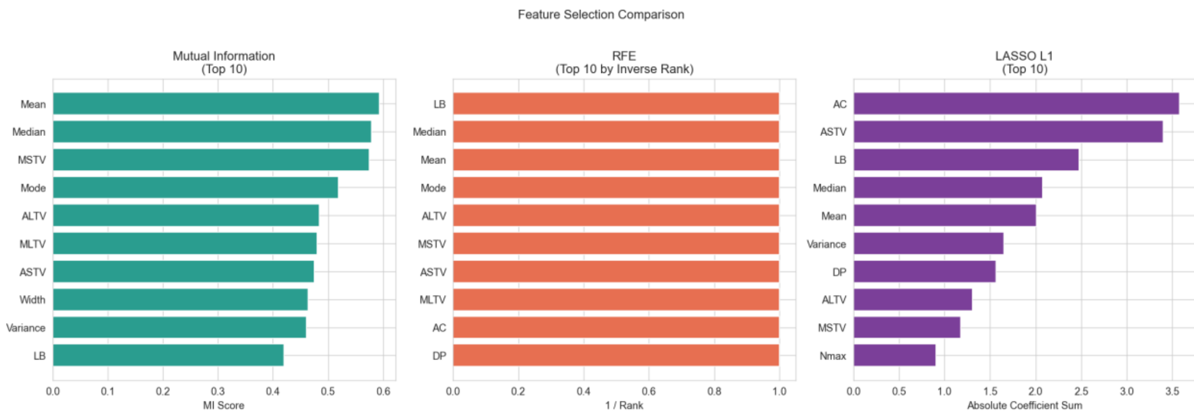

Figure 4: Top 10 features selected by ML (left), RFE (centre) and LASSO (right). Strong convergence on indices variability and central-tendency feature is evident.

The clinical significance of the consensus subset is satisfying to see in this study as ASTV, MSTV, ALTV and MLTV are direct quantifications of FHR variability and the most informative FIGO criterion [1]: absent/reduced variability is the strongest non-invasive indicator of central-nervous system hypoxia. FIGO's "reactive trace" directly encoded the AC, and the most concerning single wave feature is DP.

The baseline of FHR is simultaneously encoded by the central-tendency features like Mean, Median, Mode and LB. Hence the 11-feature subset recovers the criteria that clinicians use in practice [1].

5.3. Performance of Classifiers; Three-class Problem

Table 1 reports the performance of the test-set of all four classifiers on the 3-class problem with 21 features. In this study, XGBoost achieved the highest accuracy of **94.6%** and macro-F1 of **89.2%,** with Random Forest taking the second place with 93.4% and 88.4% accuracy and macro F1 respectively. Both methods significantly outperformed Logistic Regression of 86.9%/78.4% and SVM 89.7%/82.2% accuracy and macro F1.

The classifiers used in this study achieved a test ROC-AUC above **0.95**, decisively answering question 1 of this study: the 21 CTG features can predict fetal state far above chance.

Model	Accuracy	Macro-F1	Sensitivity	Specificity	Roc-AUC
Logistic Regression	0.869	0.784	0.842	0.937	0.956
SVM (RBF)	0.897	0.822	0.835	0.923	0.975
Random Forest	0.934	0.884	0.900	0.954	0.982
XGBoost	**0.946**	**0.892**	**0.886**	**0.948**	**0.982**

*Table 1: Test-set performance of three-class problem of all 21 features*s.

In Figure 5, the macro-F1 axis has the most sensitive performance on the uncommon Pathologic class which has the largest gap between the classifier families. This figure highlights the fact that it headlines accuracy figures alone would underestimate the benefit of the ensemble methods on the unbalanced dataset.

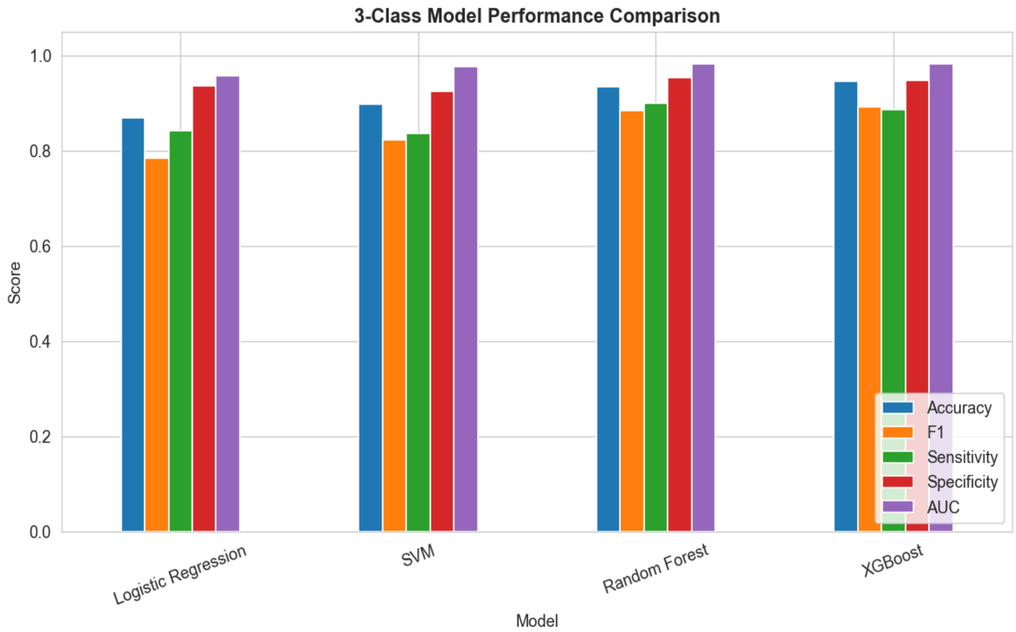

Figure 5: Bar chat of test-set metrics of the four classifiers on the three-class problem, showing the ensemble methods dominates.

5.4. Confusion-matrix Analysis

The clinically significant cell is seen at the bottom left of Figure 6 showing true Pathologic predicted Normal, such result represents a missed signal of fetal compromise.

Logistic regression and Random Forest made 1 of such error whilst SVM and XGBoost made 3 and 2 of such error respectively out of 35 pathologic cases, all being very low. The dominant residual error is Normal-Suspect confusion, this is less clinically dangerous because all Suspect cases are flagged for additional monitoring rather than an intermediate intervention [1].

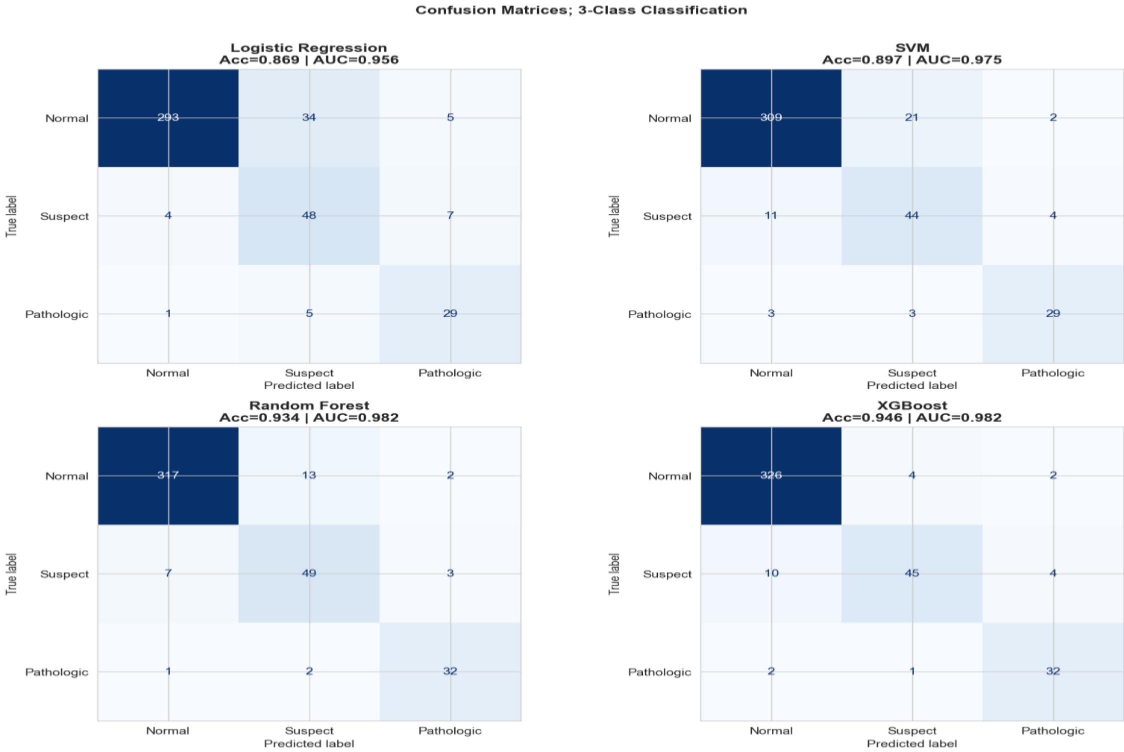

Figure 6: Confusion matrices for the four tuned classifiers on the three-class problem.

5.5. Binary Classification

In practice, collapsing the Suspect and Pathologic cases into Abnormal class is a clinically defensible simplification, as each label triggers and increased surveillance [1]. Table 2 below summarises the binary result as XGBoost leads with accuracy of **95.5%** and **AUC = 0.988**, Random Forest had a near tie. It was important to note that sensitivity for Abnormal class rises sharply (**0.93** for XGBoost vs **0.89** in the 3-class case), this indicates that a useful operating point was achieved by removing the difficult NSP boundary when the goal was to simply not flag anything not Normal.

Model	Accuracy	Macro-F1	Sensitivity	Specificity	Roc-AUC
Logistic Regression	0.880	0.848	0.904	0.947	0.954
SVM (RBF)	0.941	0.919	0.940	0.936	0.983
Random Forest	0.951	0.928	0.926	0.883	0.988
XGBoost	**0.955**	**0.935**	**0.933**	0.894	**0.988**

Table 2: Test-set performance; binary problem (Normal vs Abnormal)

Figure 7 shows the binary confusion matrices and ROC curves with AUC ≥ 0.98 by all three non-linear models further confirms that the binary problem is highly tractable.

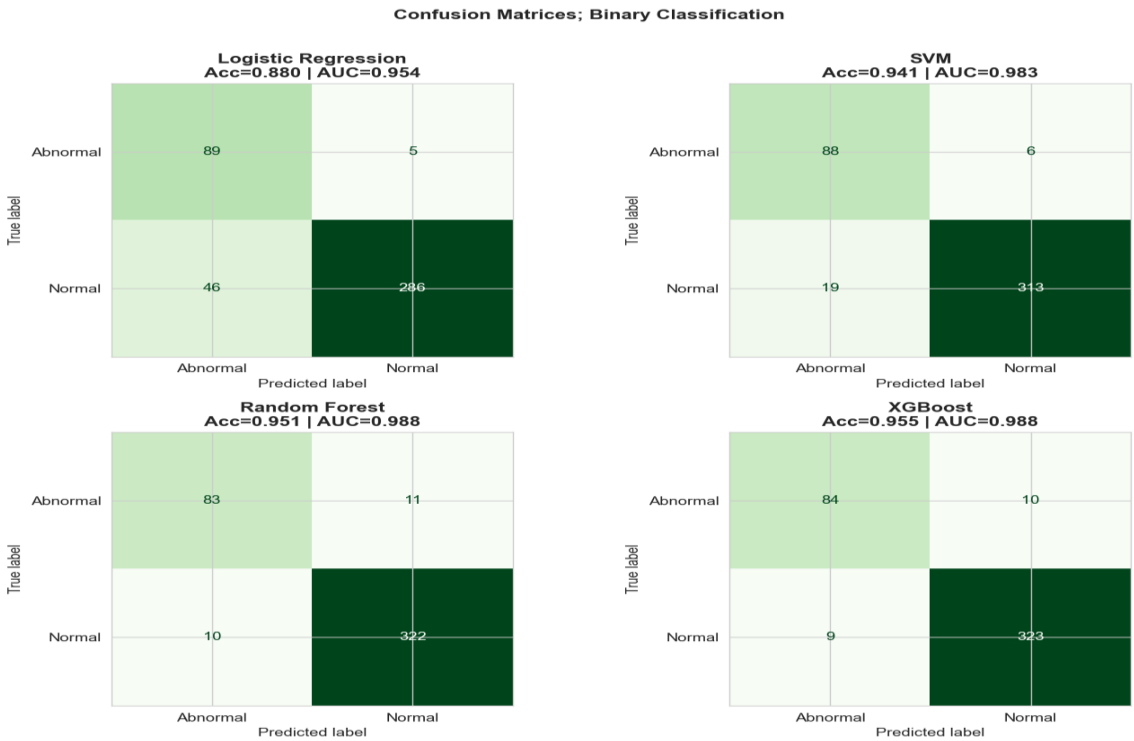

Figure 7: Confusion matrices of all four classifiers on the binary problem.

It is worth noting that, Figure 7 exposes a real trade-off. The false negative on the top right of each panel, are 5 and 6 for Logistic Regression and SVM respectively and 11 and 10 for Random Forest and XGBoost respectively. The ensemble method does not achieve the best sensitivity in this binary task; however, they win on overall accuracy because they detect more on Abnormal cases, not because of a higher specificity. In a deployed screening context, this trade-off would need to be weighed seriously since missing an Abnormal case cost more.

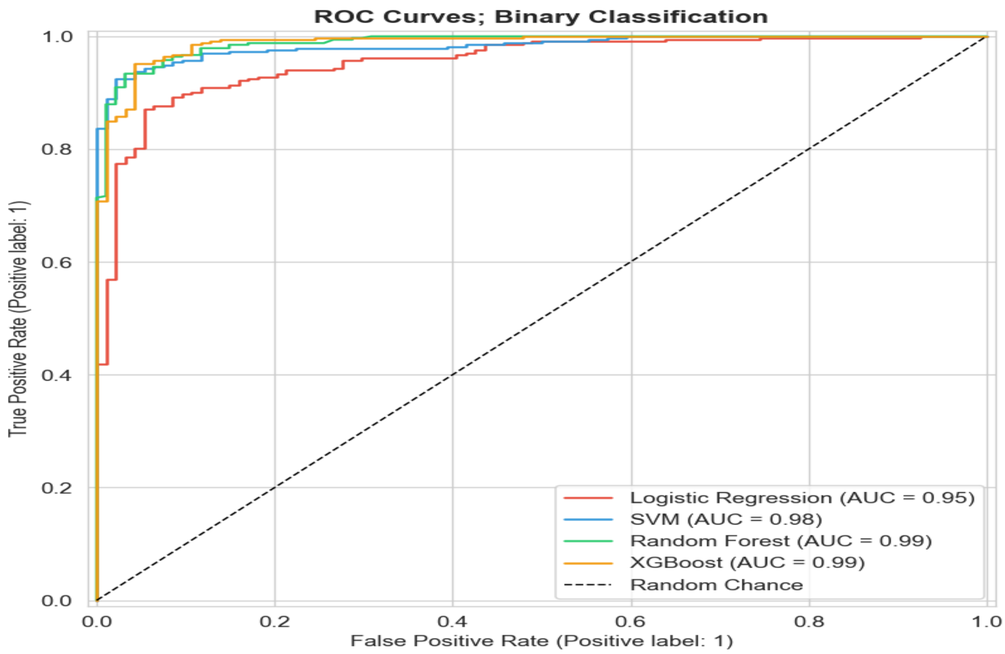

Figure 8: ROC curve for binary classification task with AUC

5.6. Reduced Feature Subset

The 3-class experiment was repeated using the 11 consensus features, which emerged from the rule defined in section 3.3, that retained any feature selected by at least 2 of the three-feature selection. 8 features were selected unanimously by the three methods and three more were selected by a least 2 methods.

This experiment using this subset yielded almost no decrease in performance for any of the classifiers as XGBoost dropped from **94.6% to 94.1%** and AUC from 0**.982 to 0.981** while maintaining the same macro F1 score. Random Forest dropped from **93.4% to 93%** accuracy and AUC from **0.982 to 0.979**. An improvement was observed in the reduced SVM with macro F1 from **0.822 to 0.858.** This result answers the question 3 if a substantially smaller feature would retain its full predictive power of the original 21 features. This is clinically significant as a model that depends on only 11 features is easier to explain and less prone to artefactual feature drift across CTG hardware vendors, more importantly easier to explain to clinicians.

5.7. Feature Importance and Most Predictive Feature

As seen in Figure 9 below, the Random Forest Gini-impurity importances, where the red bars are the consensus selected features, the dominant features(ASTV, ALTV, Mean, MSTV, AC and Median) seen are the consensus selected features, however, XGBoost produces a similar but not identical ordering with Mean, ALTV, ASTV, AC and DP sitting at the top. Both models diverge somewhat, XGBoost weighs some non-consensus features, however the reduced feature experiment in Section 5.6 confirms that this divergence does not automatically translate into meaningful predictive advantage as XGBoost has **94.1%** accuracy with the 11 consensus features.

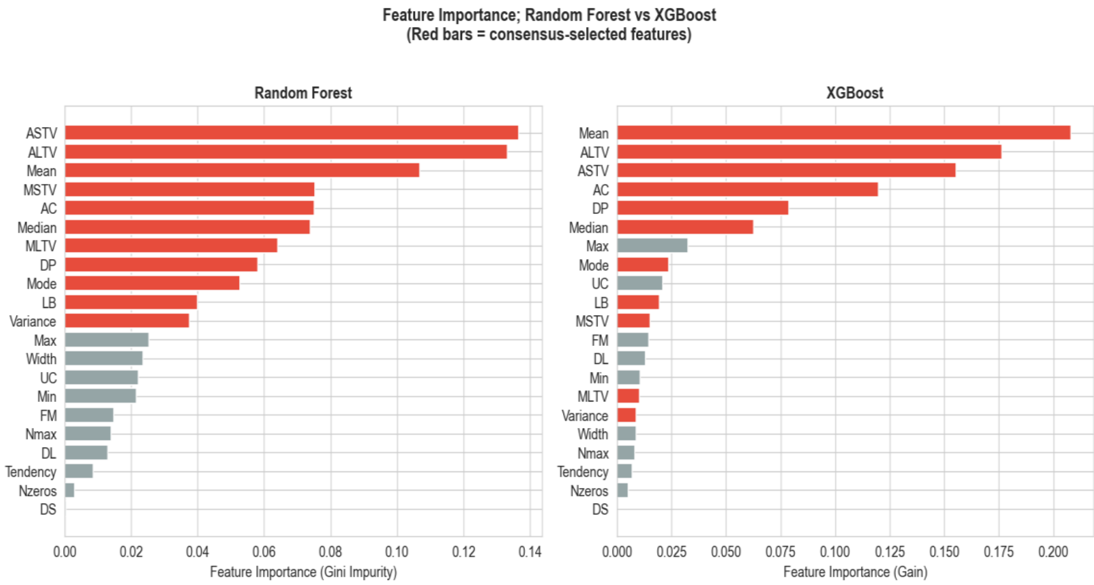

Figure 9: Feature Importance of Random Forest and XGBoost

5.8. Robustness Check

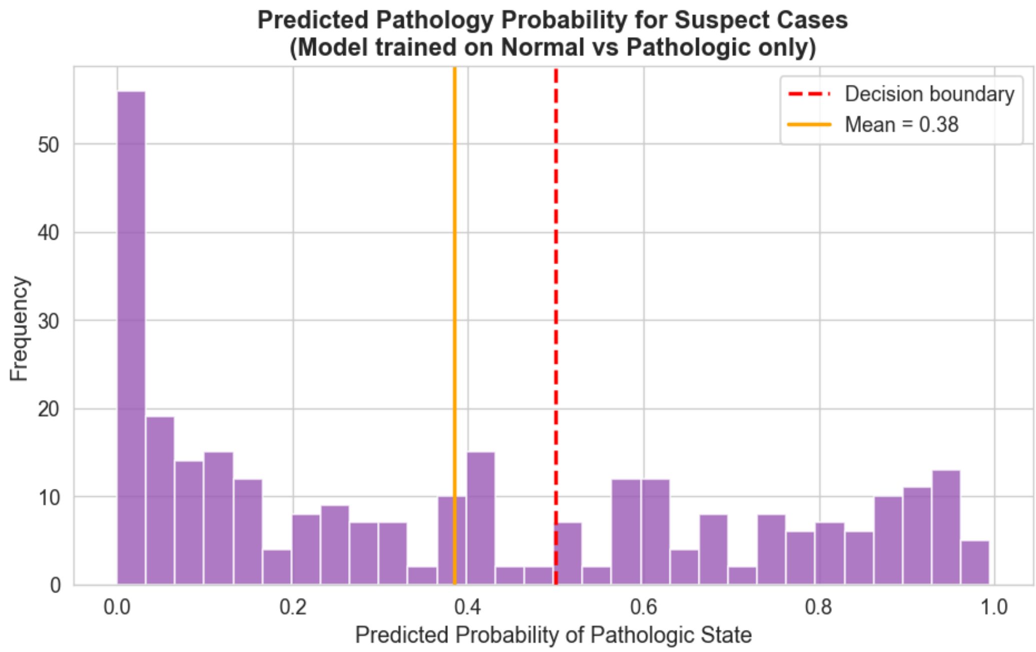

Figure 10: Predicted pathology for Suspect class, from XGBoost model trained on extreme cases

XGBoost model was trained on 1,655 Normal and 176 Pathologic cases and then applied to 295 Suspect cases. Assuming the Suspect were just simply difficult to classify as intermediate, predicted pathology probabilities for Suspect cases would be expected to cluster around 0.5. However, Figure 10 shows a clear bimodal distribution; about **38%** of Suspect cases were classified as Pathologic and **62%** as Normal. The mean predicted probability was **0.385**.

This result suggests that the Suspect class is heterogeneous, which is a subset of patients whose CTG features resembles Pathologic patients, and a larger subset resembles Normal patients. From a clinical perspective, the observation is consistent with FIGO 2015, that stated that 'suspicious' tracings are an instruction to obtain more information rather than a stable diagnosis.

5.9. Result Overview

Figure 11 displays the summary of the entire experiment; with all 12 models trained across five metrics and two patterns seem to dominate. ROC-AUC sits above 0.94 across all models in every framing which is confirmed with extensive cross-validation. This answers the first question of this coursework. Secondly, the difference between the worst performing model (Logistic Regression on 3-class reduced) and the best (XGBoost / binary) configuration is approximately 11% pp of accuracy and 17pp macro F1 respectively. This demonstrates that the choice of classifier and problem framing matters significantly even when the underlying signal is strong.

Figure 11: Heatmap of test-set performance across the 12 trained model-framing combination.

5.10. Discussion and Clinical Significance

In this study, the most clinical consequential error worth noting is the False-negative Pathologic predictions, as they represent missed fetal compromise. In the best 3-class model, only 2 of 35 cases were missed whilst in the binary setting, only 10 of 94 Abnormal cases were predicted Normal. In inter-observer studies of FIGO 2015 [1], CTG interpretations by trained midwives reported a fair agreement with the lowest being on Suspicious and Pathologic categories [17], so model error rates of this extent are competitive with established clinical disagreement. If this system was to be deployed, it would need to be more conservative than 0.5 to remove the false-negative rate close to 0.

False-positive rates are clinically less harmful however they generate avoidable interventions and consumption of resources. The XGBoost model produced only 2 errors out of 332 Normal cases, this is a rate that is favourably compared with low positive predictive values that are documented [1]. One should also note that as seen in Figure 7, simpler classifiers like LR and SVM missed fewer cases in the binary setting than the ensemble methods. This suggests that the ‘best’ model is dependent on if missed cases carry a higher clinical cost.

ML tools should be introduced as an additional input with the rest of the clinical picture rather than a replacement for expert judgement, this is a principle supported by FIGO guidelines [1] and scoping reviews [3].

5.11. Limitations.

Some limitations acknowledged in this study includes UCI dataset is small and was collected in Portugal in the 1990s at a single site, hence figures cannot assume to capture other populations or modern CTG hardware [3]. Secondly, although SMOTE was applied only to the training partition, the generated synthetic samples were drawn from the same source of distribution and would not correct sampling bias in the original cohort. Lastly, hyperparameter grids were modest, and the cross-validation was only performed on a single seed, therefore a larger study should incorporate its variability across multiple splits and seed.

6. CONCLUSION

A complete reproducible machine learning pipeline for fetal-state classification was delivered in this piece of work. More importantly, the three coursework questions were answered. Firstly, the 21 quantitative CTG features can predict fetal state with a very high accuracy: the best 3-class model (XGBoost) achieved 94.6% accuracy and AUC = 0.982 whilst the best binary model achieved AUC = 0.988 which is far above the 0.50 chance threshold, also above the level of agreement reported among independent expert obstetricians on the same trace. Secondly, the most predictive features are the FHR variability indices (ASTV, ALTV, MSTV, MLTV), FHR central tendency (Mean, Median, Mode, baseline LB), number of accelerations (AC) and the count of prolonged decelerations (DP). This ranking was agreed upon by Mutual Information, RFE and LASSO, which mirrors FIGO 2015 clinical interpretation criteria. Thirdly, the 11-feature consensus subset retains its full predictive performance, showing that a compact clinically interpretable model is achievable.

Beyond these core questions, the robustness was explored, showing that the Suspect class is bimodal under a Normal vs Pathologic model suggesting heterogeneity rather than it being a uniform intermediate. Across the three ensemble methods used in this study Random Forest, XGBoost clearly outperformed Logistic Regression and SVM and the choice of the problem; binary vs 3-class and where the decision threshold matter as much as the choice of model.

External validation on contemporary multi-site cohort with hard fetal outcome should be explored and prospective evaluation against expert obstetrician as well as integration with explainable tools like SHAP to surface case-level rationales. These should be the next steps before practical translation of these results. Until these steps are taken, these present models should be regarded as methodological demonstration rather than a clinically validated tool.

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8. APPENDIX

Student Declaration of AI Tool use in this Assessment

Solo Work	**S1 - Generative AI tools have not been used for this assessment.**	☐
Assisted Work	**A1 – Idea Generation and Problem Exploration** Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.	☐
	A2 - Planning & Structuring Projects AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student’s own work.	☐
	A3 – Code Architecture AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student’s own work.	☐
	A4 – Research Assistance	☐

	Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions. The interpretation and integration of research into the assignment remain the student's responsibility.	
	A5 - Language Refinement Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.	☒
	A6 – Code Review AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.	☒
	A7 - Code Generation for Learning Purposes Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.	☐
	A8 - Technical Guidance & Debugging Support AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.	☒
	A9 - Testing and Validation Support AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.	☐
	A10 - Data Analysis and Visualization Guidance AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.	☐
	A11 - Other uses not listed above Please specify:	☐
Partnered Work	**P1 - Generative AI tool usage has been used integrally for this assessment** Students can adopt approaches that are compliant with instructions in the assessment brief. Please Specify:	☐

Please provide details of AI usage and which elements of the coursework this relates to:

I understand that the ownership and responsibility for the academic integrity of this submitted assessment falls with me, the student.	☒
I confirm that all details provide above are an accurate description of how AI was used for this assessment.	☒